

MiniGPT: 2-Month Progress Report

Comparison: 2026-01-19 vs 2026-03-29

Metric	Value
Lines of Code Added	4,620+
Lines of Code Removed	41,370-
Net Code Reduction	86%
Training Speedup	10x
GPU Throughput Gain	2-4x

Key Achievements

- ✓ 10x training speedup (KV cache bypass + FP16 + fused kernels)
- ✓ FP16 mixed precision (Tensor Core activation on RTX 3060)
- ✓ Fused CUDA kernels reducing launch overhead
- ✓ FineWeb-Edu streaming pipeline (10GB real-world data)
- ✓ Complete gradient validation system
- ✓ VRAM budgeting utilities
- ✓ Production-ready training loop with checkpointing
- ✓ BPE tokenizer with GPT-4 regex

Technical Improvements

GPU Backend Abstraction: Unified CuPy/NumPy interface with VRAM monitoring, FP16 matmul, fused kernel support. Impact: 2-4x throughput

FP16 Mixed Precision: Forward in FP16 (Tensor Cores), backward in FP32. Auto-enabled on GPU. Impact: 2-4x matmul speedup

Fused CUDA Kernels: `_fused_silu`, `_fused_silu_backward`, `_fused_exp` reduce kernel launches. Impact: ~36 fewer launches/step

KV Cache Bypass Training: Skip per-token loop during training. 10x speedup (0.3s → 0.03s). Cache used only for inference.

Real-World Data: FineWeb-Edu streaming (10GB, 2B tokens). Replaced synthetic TinyStories.

Gradient Validation: Central finite differences with float64 loss. Catches bugs before training.

VRAM Budgeting: Component-wise memory estimation. Fits 12GB RTX 3060 with T=512, B=128.

Training Performance Comparison

Metric	Old (2026-01-19)	New (2026-03-29)
CPU (NumPy)	~30 sec/step	~30 sec/step
GPU (FP32)	N/A	~0.3 sec/step
GPU (FP16)	N/A	~0.1-0.15 sec/step
Data	TinyStories (75KB)	FineWeb-Edu (10GB)

72-Hour Training (50K steps)

- FP32: ~4.2 hours
- FP16: ~1.4-2.1 hours (projected)
- VRAM: ~6.8 GB / 12 GB RTX 3060

Code Quality & Structure

New src/minigpt/ Structure

backend.py: GPU/CPU abstraction, VRAM monitoring, FP16 matmul, fuse decorator

model.py: Transformer architecture with fused kernels

config.py: Configuration management

optimizer.py: AdamW with custom weight decay

tokenizer.py: BPE tokenizer with GPT-4 regex

inference.py: Generation with nucleus sampling

Documentation

- CLAUDE.md: 220+ line project context for future sessions
- ROADMAP_72H.md: 72-hour training schedule with milestones
- README.md: Project overview and quickstart

Summary & Status

The 2-month refactor transformed MiniGPT from a basic transformer experiment into a production-ready training system optimized for consumer GPUs.

Ready for 72-Hour Training Event ✓

- ✓ All gradient bugs fixed and validated
- ✓ FP16 mixed precision activates Tensor Cores (2-4x speedup)
- ✓ Fused CUDA kernels reduce kernel launch overhead
- ✓ FineWeb-Edu pipeline tested end-to-end
- ✓ VRAM budgeted for 12GB GPU
- ✓ Expected result: ~PPL 15 on real-world text

Generated: 2026-03-29 23:49:48 | Repository: MiniGPT | Branch: main

GPT-1 Roadmap: Progress Timeline

Extension to the 2026-03-29 progress report. This section documents the pivot from the Llama-3-style 12.9M configuration to a faithful GPT-1 (Radford et al., 2018) replica at 117M parameters, on the same RTX 3060 12 GB target hardware.

1. Milestone Timeline

Date	Milestone	Scope & Expected Duration
2026-03-29	M0 — Production Foundation (complete)	FP16 Tensor Cores, fused CuPy kernels, FineWeb-Edu pipeline, gradient validation
2026-04-17 → +3 days	M1 — Architecture Retrofit	Replace RMSNorm → LayerNorm (gamma + beta, eps=1e-5, post-norm). Replace S
+3 → +5 days	M2 — Optimizer & Regularization	Replace AdamW (decoupled) → Adam + coupled L2 weight decay. Hyperparameter
+5 → +10 days	M3 — Tokenizer & Embedding	Upgrade BPE tokenizer to 40,000 vocab using HuggingFace `tokenizers` Rust imple
+10 → +17 days	M4 — Scale to 117M (GPT-1 Target)	model=768, n_layers=12, n_heads=12, d_ff=3072, max_len=512. VRAM validation

2. Current MiniGPT vs. Target GPT-1

Snapshot of architectural parameters. The "Current" column reflects the working tree on 2026-04-17: model.py and optimizer.py contain uncommitted GPT-1 refactors; config.py, tokenizer.py, and trainer.py still hold legacy values.

Dimension	Current (2026-04-17)	Target GPT-1
Parameters	~14.2M (uncommitted)	~117M
Layers (n_layers)	6	12
Hidden dim (d_model)	384	768
Attention heads	6 Q / 2 KV (GQA)	12 / 12 (full MHA)
FFN hidden (d_ff)	1024 (SwiGLU ratio)	3072 (4 × d_model)
Normalization	LayerNorm post-norm *	LayerNorm post-norm
Activation	GELU tanh-approx *	GELU tanh-approx
Positional encoding	Learned absolute W_pos *	Learned absolute
Output projection	Tied to W_emb.T *	Tied to W_emb.T
Optimizer	Adam + coupled L2 *	Adam + L2
LR / betas / eps	2.5e-4 / (0.9, 0.98) / 1e-8 *	2.5e-4 / (0.9, 0.98) / 1e-8
Weight decay (1D)	LN γ/β , biases excl. *	LN γ/β , biases, W_pos excl.
Dropout	wired at 4 sites, p=0.0 †	wired at 4 sites, p=0.1
Vocab size	4,096 BPE (only <EOS>)	40,000 BPE (pad/eos/unk)
Context length	512	512

Dataset	FineWeb-Edu (10 GB stream)	FineWeb-Edu (10 GB stream)
---------	----------------------------	----------------------------

* Implemented in the uncommitted model.py / optimizer.py refactor. † Dropout plumbing exists; p must be set to 0.1 via config default update.

3. VRAM Projection: 117M on RTX 3060 12 GB

Component-wise memory budget for the full GPT-1 configuration (d=768, L=12, H=12, V=40,000, T=512). Compared against the estimate_model_vram() formula in backend.py (with FFN ratio updated from SwiGLU 2·4d/3 to GPT-1 4d). Batch × accum sized for effective batch 64.

Component	FP32 (all paths)	FP16 mixed (fwd matmul)
Model params (master, FP32)	468 MB	468 MB
Optimizer (m + v, FP32)	936 MB	936 MB
Gradients (FP32)	468 MB	468 MB
Activations (12 L, T=512)	~9,100 MB	~4,550 MB
■ attn scores B·H·T·T	~1,500 MB @ B=8	~750 MB @ B=8
■ FFN intermediate	~4,600 MB @ B=8	~2,300 MB @ B=8
■ layer I/O (B·T·D)	~3,000 MB @ B=8	~1,500 MB @ B=8
FP16 temporaries	—	~300 MB
CUDA + allocator overhead	~500 MB	~500 MB
Total	~11,500 MB	~7,200 MB
Headroom (12,288 MB GPU)	~780 MB (tight)	~5,080 MB (comfortable)

4. Projected Batch Configurations & Training Time

Precision	Micro batch	Accum	Effective batch	Peak VRAM	Time/step (est.)	50K steps
FP32	4	16	64	~9.7 GB	~1.0 s	~14 h
FP32	8	8	64	~11.5 GB	~0.9 s	~12 h (tight)
FP16 mixed	8	8	64	~7.2 GB	~0.35 s	~4.9 h
FP16 mixed	16	4	64	~9.5 GB	~0.30 s	~4.2 h
FP16 mixed	24	4	96	~11.2 GB	~0.40 s	~5.6 h

Estimates assume: Tensor Core utilization at 60% (RTX 3060 is Ampere SM86), PCIe bandwidth non-limiting, cupy.fuse'd element-wise kernels, KV-cache bypass during training (already implemented). Benchmark with scripts/benchmark_gpu.py before committing to a final config.

5. Exit Criteria for GPT-1 Milestone

- Architecture matches GPT-1 paper section 3.1: LayerNorm post-norm, GELU, learned positional embeddings, tied output, 12 layers of 768 dim × 12 heads.
- Optimizer matches paper section 4: Adam, lr peaks at 2.5e-4 over 2000 warmup steps, cosine decay thereafter, batch size 64 × seq 512.
- Tokenizer at 40,000 BPE with , , special tokens; FineWeb-Edu shards re-encoded.
- Model instantiates at 117M ± 2M parameters. Tied embedding verified to appear exactly once in named_parameters() (no duplicate output head).
- Step-0 validation loss $\approx \ln(40000) = 10.60 \pm 0.1$ (random init sanity).
- 100-step smoke training run: loss monotonically decreases, no NaN/Inf, grad norm in [0.1, 5.0] post-warmup.
- Peak measured VRAM ≤ 11.0 GB at chosen batch × accum configuration.
- scripts/validate_gradients.py: median relative error <10%, p90 <50% on new modules (LayerNorm, GELU FFN, learned-pos embedding).